

Practice problems #14

1. Let

$$A = \begin{bmatrix} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{bmatrix}.$$

- (a) Compute AB using row-column rule.
- (b) Compute AB using column-row expansion.
- (c) Let A and B be decomposed as follows, where A_{11} is of size 2×2 . Compute AB using block matrices.

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}.$$

2. Suppose that A_{11} is an invertible matrix. Find matrices X and Y such that the product below has the form indicated. Also, compute B_{22} .

$$\begin{bmatrix} I & 0 & 0 \\ X & I & 0 \\ Y & 0 & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix} = \begin{bmatrix} B_{11} & B_{12} \\ 0 & B_{22} \\ 0 & B_{32} \end{bmatrix}.$$

3. Suppose A_{11} is invertible. Find X and Y so that

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I & 0 \\ X & I \end{bmatrix} \begin{bmatrix} A_{11} & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I & Y \\ 0 & I \end{bmatrix}$$

where $S = A_{22} - A_{21}A_{11}^{-1}A_{12}$. The matrix S is called the *Schur complement* of A_{11} . Likewise, if A_{22} is invertible, the matrix $A_{11} - A_{12}A_{22}^{-1}A_{21}$ is called the Schur complement of A_{22} . Such expression occur frequently in the theory of systems engineering, and elsewhere (don't need to memorize).

Note that this is a factorization of A into block diagonal and block triangular matrices.

Answers or Hints are on page 2.

#1: Do them! You should get same answer for all.

#2: $X = -A_{21}A_{11}^{-1}, \quad Y = -A_{31}A_{11}^{-1}, \quad B_{22} = A_{22} - A_{21}A_{11}^{-1}A_{12}$

#3: $X = A_{21}A_{11}^{-1}, \quad Y = A_{11}^{-1}A_{12}$